Lab – File System Commands

Introduction

In this lab, you will use CLI commands to manage files and folders in Windows.

Recommended Equipment

- A computer running Windows

Instructions

Step 1: Access the Windows command prompt.

a. Log on to a Windows computer. The account **ITEUser** is used as the example user account throughout this lab.

b. To access the Windows command prompt, click **Start** and enter **cmd**. Select **Command Prompt** or **cmd.exe** to continue.

Step 2: Create and change directories.

In this step, you will use the change directory (**cd**), make directory (**md**), and directory (**dir**) commands.

Note: A directory is another word for folder. Directory and folder are used interchangeably throughout this lab.

a. Type **cd** at the command prompt.

Question:

What is the current directory?

```
C:\Users\TIPQC>cd
C:\Users\TIPQC
```

b. Type **dir** at the command prompt to list the files and folders that are in the current folder.

```
C:\Users\ITEUser> dir
 Volume in drive C has no label.
 Volume Serial Number is 9055-35E9

 Directory of C:\Users\ITEUser

04/27/2019  09:21 AM    <DIR>          .
04/27/2019  09:21 AM    <DIR>          ..
04/25/2019  11:39 AM    <DIR>          3D Objects
04/25/2019  11:39 AM    <DIR>          Contacts
04/26/2019  10:29 AM    <DIR>          Desktop
04/25/2019  11:39 AM    <DIR>          Documents
04/25/2019  11:39 AM    <DIR>          Downloads
<some output omitted>
               0 File(s)              0 bytes
              15 Dir(s)  32,671,969,280 bytes free
```

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```
Microsoft Windows [Version 10.0.26100.8246]
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C:\Users\TIPQC>dir
 Volume in drive C is Q3203-16
 Volume Serial Number is 7029-55C0

 Directory of C:\Users\TIPQC

10/07/2026  07:55 PM    <DIR>          .
03/09/2026  11:03 AM    <DIR>          ..
07/30/2025  06:05 PM    <DIR>          .anaconda
01/21/2025  10:55 AM    <DIR>          .azuredatastudio
10/02/2025  04:44 PM    <DIR>          .cache
09/24/2025  05:22 PM    <DIR>          .conda
07/30/2025  06:05 PM                25 .condarc
07/30/2025  06:05 PM    <DIR>          .continuum
08/01/2026  03:08 PM    <DIR>          .copilot
07/30/2025  06:10 PM    <DIR>          .ipython
07/30/2025  06:13 PM    <DIR>          .jupyter
07/30/2025  06:10 PM    <DIR>          .matplotlib
09/03/2026  12:09 PM               176 .packettracer
07/18/2026  02:35 PM    <DIR>          .spyder-py3
10/08/2026  11:20 AM    <DIR>          .VirtualBox
08/06/2025  05:06 PM    <DIR>          .vscode
07/18/2026  04:41 PM    <DIR>          .vscode-shared
09/03/2026  01:33 PM    <DIR>          Cisco Packet Tracer 8.2.2
10/18/2024  05:21 AM    <DIR>          Contacts
04/28/2026  09:15 AM    <DIR>          Desktop
09/18/2026  06:27 PM    <DIR>          Documents
10/08/2026  11:05 AM    <DIR>          Downloads
05/02/2025  08:23 AM    <DIR>          Favorites
02/10/2026  10:59 AM    <DIR>          flutter
08/06/2025  06:50 PM             2,361 from PyQt6.py
07/30/2025  06:34 PM    <DIR>          Jedi
10/18/2024  05:21 AM    <DIR>          Links
10/07/2026  05:26 PM    <DIR>          Microsoft
10/18/2024  05:21 AM    <DIR>          Music
10/18/2024  05:23 AM    <DIR>          OneDrive
09/16/2026  02:38 PM    <DIR>          Pictures
08/01/2026  03:09 PM    <DIR>          PyQt6
10/18/2024  05:21 AM    <DIR>          Saved Games
10/17/2024  03:58 PM    <DIR>          Searches
08/06/2025  05:09 PM    <DIR>          venv
04/14/2026  09:59 AM    <DIR>          VHDPlus
05/21/2026  02:10 PM    <DIR>          Videos
10/08/2026  11:20 AM    <DIR>          VirtualBox VMs
               3 File(s)          2,562 bytes
              35 Dir(s)  526,138,740,736 bytes free

C:\Users\TIPQC>md.BossBlazo

C:\Users\TIPQC>
```

c. In the current directory, use the **md** command to create three new folders: **ITEfolder1**, **ITEfolder2**, and **ITEfolder3**. Type **md ITEfolder1** and press **Enter**. Create **ITEfolder2** and **ITEfolder3**. (**Note**: The command **mkdir** performs the same function as the command **md**.)

```
C:\Users\ITEUser> md ITEfolder1
C:\Users\ITEUser> md ITEfolder2 ITEfolder3
```

```
10/08/2026  11:24 AM    <DIR>          .BossBlazo
10/08/2026  11:41 AM    <DIR>          .BossBlazo2
10/08/2026  11:41 AM    <DIR>          .BossBlazo3
```

d. Type **dir** to verify that the folders have been created.

e. Type **cd ITEfolder3** at the command prompt and press **Enter**.

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Question:

Which folder are you in now?

```
C:\Users\TIPQC>cd .BossBlazo3

C:\Users\TIPQC\.BossBlazo3>
```

f. Create **ITEfolder4** in the current directory. Within the **ITEfolder4** folder, create a folder named **ITEfolder5**. Use the **dir** command to verify the folder creation.

Question:

What command or commands did you use to create the nested folders?

```
C:\Users\TIPQC\.BossBlazo3\.BossBlazo4>dir
 Volume in drive C is Q3203-16
 Volume Serial Number is 7029-55C0

 Directory of C:\Users\TIPQC\.BossBlazo3\.BossBlazo4

10/08/2026  11:52 AM    <DIR>          .
10/08/2026  11:51 AM    <DIR>          ..
10/08/2026  11:52 AM    <DIR>          .BossBlazo5
               0 File(s)              0 bytes
               3 Dir(s)  522,126,536,704 bytes free
```

g. Change directory as necessary until you are in ITEfolder5.

```
C:\Users\TIPQC\.BossBlazo3\.BossBlazo4>cd .BossBlazo5
```

h. Type **cd ..** to change the current directory. Each **..** is a shortcut to move up one level in the directory tree.

Question:

After issuing the **cd ..** command, what is your directory now?

```
C:\Users\TIPQC\.BossBlazo3\.BossBlazo4\.BossBlazo5>cd ..
```

What would be the current directory if you issue this command at C:\Users\ITEUser\ITEfolder3\ITEFolder4?

```
C:\Users\TIPQC\.BossBlazo3\.BossBlazo4>
```

Step 3: Create text files.

a. Navigate to the **C:\Users\ITEUser\ITEfolder1** directory. Type **cd ..\ITEfolder1** at the prompt.

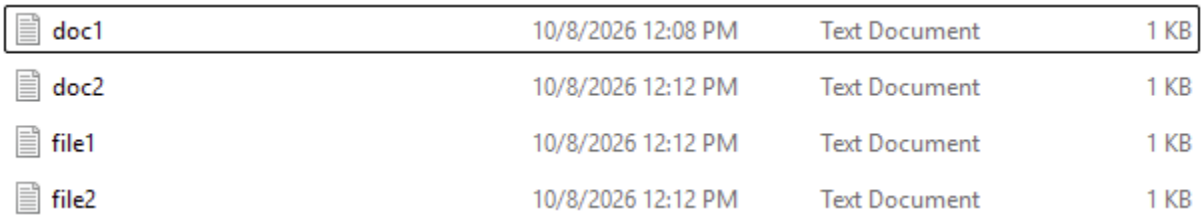
```
C:\Users\TIPQC\.BossBlazo>
```

b. Type **echo This is doc1.txt > doc1.txt** at the command prompt. The **echo** command is used to display a message at the command prompt. The **>** is used to redirect the message from the screen to a file. For example, in the first line, the message **This is doc1.txt** is redirected into a new file named **doc1.txt**. Use the **echo** command and **>** redirect to create these files: **doc2.txt**, **file1.txt**, and **file2.txt**.

```
C:\Users\ITEUser\ITEfolder1> echo This is doc1.txt > doc1.txt
```

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```
C:\Users\ITEUser\ITEfolder1> echo This is doc2.txt > doc2.txt
C:\Users\ITEUser\ITEfolder1> echo This is file1.txt > file1.txt
C:\Users\ITEUser\ITEfolder1> echo This is file2.txt > file2.txt
```



c. Use the **dir** command to verify the files are in the **ITEfolder1** folder.

```
C:\Users\ITEUser\ITEFolder1> dir
<some output omitted>
04/29/2019  08:05 AM                19 doc1.txt
04/29/2019  08:06 AM                19 doc2.txt
04/29/2019  08:08 AM                20 file1.txt
04/29/2019  08:08 AM                20 file2.txt
               4 File(s)             78 bytes
               2 Dir(s)  32,625,397,760 bytes free
```

```
C:\Users\TIPQC\.BossBlazo>dir
 Volume in drive C is Q3203-16
 Volume Serial Number is 7029-55C0

 Directory of C:\Users\TIPQC\.BossBlazo

10/08/2026  12:12 PM    <DIR>          .
10/08/2026  11:41 AM    <DIR>          ..
10/08/2026  12:08 PM                19 doc1.txt
10/08/2026  12:12 PM                19 doc2.txt
10/08/2026  12:12 PM                20 file1.txt
10/08/2026  12:12 PM                20 file2.txt
               4 File(s)             78 bytes
               2 Dir(s)  520,928,423,936 bytes free
```

d. Use the **more** or **type** command to view the content of the newly created text files.

```
C:\Users\ITEUser\ITEfolder1> more doc1.txt
This is doc1.txt

C:\Users\ITEUser\ITEfolder1> type doc2.txt
This is doc2.txt
```

```
C:\Users\TIPQC\.BossBlazo>more doc1.txt
This is doc1.txt

C:\Users\TIPQC\.BossBlazo>type doc2.txt
This is doc2.txt
```

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Step 4: Copy, delete, and move files.

a. At the command prompt, type **move doc2.txt C:\Users\ITEUser\ITEfolder2** to move the file **doc2.txt** to the **C:\Users\ITEUser\ITEfolder2** directory.

```
C:\Users\ITEUser\ITEfolder1> move doc2.txt C:\Users\ITEUser\ITEfolder2
        1 file(s) moved.
```

```
C:\Users\TIPQC\.BossBlazo>move doc2.txt C:\Users\TIPQC\.BossBlazo2
        1 file(s) moved.
```

b. Type **dir** at the prompt to verify that **doc2.txt** is no longer in the current directory.

```
C:\Users\TIPQC\.BossBlazo>dir
 Volume in drive C is Q3203-16
 Volume Serial Number is 7029-55C0

 Directory of C:\Users\TIPQC\.BossBlazo

10/08/2026  12:22 PM    <DIR>          .
10/08/2026  11:41 AM    <DIR>          ..
10/08/2026  12:08 PM                19 doc1.txt
10/08/2026  12:12 PM                20 file1.txt
10/08/2026  12:12 PM                20 file2.txt
               3 File(s)             59 bytes
               2 Dir(s)  520,910,618,624 bytes free
```

c. Navigate to **C:\Users\ITEUser\ITEfolder2** to change the directory to **ITEfolder2**. Type **dir** at the prompt to verify **doc2.txt** has been moved.

```
C:\Users\TIPQC\.BossBlazo2>dir
 Volume in drive C is Q3203-16
 Volume Serial Number is 7029-55C0

 Directory of C:\Users\TIPQC\.BossBlazo2

10/08/2026  12:22 PM    <DIR>          .
10/08/2026  11:41 AM    <DIR>          ..
10/08/2026  12:12 PM                19 doc2.txt
               1 File(s)             19 bytes
               2 Dir(s)  520,669,859,840 bytes free
```

d. Type **copy doc2.txt doc2_copy.txt** to create a copy of **doc2.txt**. Type **dir** at the prompt to verify a copy of the file has been created.

```
C:\Users\ITEUser\ITEfolder2> dir
<some output omitted>
Directory of C:\Users\ITEUser\ITEfolder2

04/30/2019  09:07 AM                19 doc2.txt
04/30/2019  09:07 AM                19 doc2_copy.txt
               2 File(s)             38 bytes
               2 Dir(s)  31,753,068,544 bytes free
```

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```
C:\Users\TIPQC\.BossBlazo2>copy doc2.txt doc2_copy.txt
        1 file(s) copied.

C:\Users\TIPQC\.BossBlazo2>dir
 Volume in drive C is Q3203-16
 Volume Serial Number is 7029-55C0

 Directory of C:\Users\TIPQC\.BossBlazo2

10/08/2026  12:28 PM    <DIR>          .
10/08/2026  11:41 AM    <DIR>          ..
10/08/2026  12:12 PM                19 doc2.txt
10/08/2026  12:12 PM                19 doc2_copy.txt
               2 File(s)             38 bytes
               2 Dir(s)  520,669,700,096 bytes free
```

e. Use the **move** command to move **doc2_copy.txt** to **ITEfolder1**. Type **move doc2_copy.txt ..\ITEfolder1**.

```
C:\Users\ITEUser\ITEfolder2> move doc2_copy.txt ..\ITEfolder1
        1 file(s) moved.
```

```
C:\Users\TIPQC\.BossBlazo2>move doc2_copy.txt ..\.BossBlazo
        1 file(s) moved.
```

f. A copy of the file **doc2.txt** can be created in a new location using the **copy** command. At the prompt, enter the **copy doc2.txt ..\ITEfolder1\doc2_new.txt**.

```
C:\Users\ITEUser\ITEfolder2> copy doc2.txt ..\ITEfolder1\doc2_new.txt
        1 file(s) copied.
```

```
C:\Users\TIPQC\.BossBlazo2>copy doc2.txt ..\.BossBlazo\doc2_new.txt.
        1 file(s) copied.
```

g. The file **doc2.txt** can also be moved to a new location with a new filename using the **move** command. Type **move doc2.txt ..\ITEfolder1\doc2_new.txt** at the prompt.

```
C:\Users\ITEUser\ITEfolder2> move doc2.txt ..\ITEfolder1\doc2_move.txt
        1 file(s) moved.
```

```
C:\Users\TIPQC\.BossBlazo2> move doc2.txt ..\.BossBlazo\doc2_new.txt
Overwrite C:\Users\TIPQC\.BossBlazo\doc2_new.txt? (Yes/No/All): Yes
        1 file(s) moved.
```

h. Type **dir ..\ITEfolder1** to view the content in **ITEfolder1** without leaving the current directory.

```
C:\Users\ITEUser\ITEfolder2> dir ..\ITEfolder1
<some output omitted>
 Directory of C:\Users\ITEUser\ITEfolder1

04/29/2019  12:08 PM    <DIR>          .
04/29/2019  12:08 PM    <DIR>          ..
04/29/2019  08:05 AM                19 doc1.txt
04/29/2019  08:06 AM                19 doc2_copy.txt
04/29/2019  08:06 AM                19 doc2_move.txt
```

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```
04/29/2019  08:06 AM                19 doc2_new.txt
04/29/2019  08:08 AM                20 file1.txt
04/29/2019  08:08 AM                20 file2.txt
               6 File(s)            116 bytes
               2 Dir(s)  31,467,700,224 bytes free
```

```
C:\Users\TIPQC\.BossBlazo2>dir ..\.BossBlazo
 Volume in drive C is Q3203-16
 Volume Serial Number is 7029-55C0

 Directory of C:\Users\TIPQC\.BossBlazo

10/08/2026  12:35 PM    <DIR>          .
10/08/2026  11:41 AM    <DIR>          ..
10/08/2026  12:08 PM                19 doc1.txt
10/08/2026  12:12 PM                19 doc2_copy.txt
10/08/2026  12:12 PM                19 doc2_new.txt
10/08/2026  12:12 PM                20 file1.txt
10/08/2026  12:12 PM                20 file2.txt
               5 File(s)             97 bytes
               2 Dir(s)  520,099,135,488 bytes free
```

i. Change the current directory to **ITEfolder1**. Type **cd ..\ITEfolder1** at the prompt.

```
C:\Users\TIPQC\.BossBlazo2>cd ..\.BossBlazo
```

j. Move **file1.txt** and **file2.txt** into **ITEfolder3**. To move all the files that contain the word **file** into **ITEfolder3** with one command, use a **wildcard** (*) character to represent one or more characters. Type **move file* ..\ITEfolder3**.

```
C:\Users\ITEUser\ITEfolder1> move file* ..\ITEfolder3
C:\Users\ITEUser\ITEfolder1\file1.txt
C:\Users\ITEUser\ITEfolder1\file2.txt
        2 file(s) moved.
```

```
C:\Users\TIPQC\.BossBlazo>move file* ..\.BossBlazo3
C:\Users\TIPQC\.BossBlazo\file1.txt
C:\Users\TIPQC\.BossBlazo\file2.txt
        2 file(s) moved.
```

k. To view the content of the file doc2_new.txt, use the **type** or **more** command at the prompt.

```
C:\Users\ITEUser\ITEfolder1> type doc2_new.txt
This is doc2.txt
```

```
C:\Users\TIPQC\.BossBlazo>type doc2_new.txt
This is doc2.txt
```

l. Use the **ren** command to rename doc2_new.txt to doc3.txt. Use the **dir** command to display the content in the directory.

```
c:\Users\ITEUser\ITEfolder1> ren doc2_new.txt file.log
```

```
C:\Users\TIPQC\.BossBlazo>ren doc2_new.txt file.log
```

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m. Use the **type** or **more** command to view the content of the newly rename file **file.log.**

```
C:\Users\ITEUser\ITEfolder1> more file.log
This is doc2.txt
```



n. To delete a file or multiple files, use the **del** command with the file names at the prompt.

Question:

What single command would you use to delete all the files with doc2 in the filename? Use the command to remove the files with doc2 in the filename.

del *doc2*

What command would you use to delete all the files in the directory? Use the command to remove the files.

del *.*

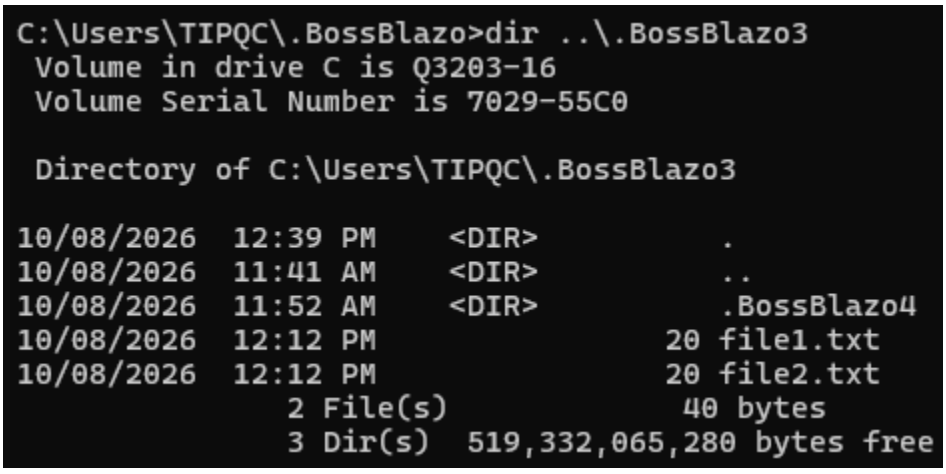
Step 5: Use the xcopy and robocopy commands.

In this step, the **xcopy** and **robocopy** commands are used to copy the content in a directory.

a. View the content of ITEfolder3.

```
C:\Users\ITEUser\ITEfolder1> dir ..\ITEfolder3
<some output omitted>
 Directory of c:\Users\ITEUser\ITEfolder3

04/29/2019  03:11 PM    <DIR>          .
04/29/2019  03:11 PM    <DIR>          ..
04/29/2019  03:05 PM                20 file1.txt
04/29/2019  03:05 PM                20 file2.txt
04/29/2019  03:01 PM    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  31,492,157,440 bytes free
```



b. Type **xcopy ..\ITEfolder3 .** at the prompt to copy the content of **ITEfolder3** to **ITEfolder1**. Note the **.** at the end of the command. It is a shortcut for the current directory.

```
C:\Users\ITEUser\ITEfolder1> xcopy ..\ITEfolder3 .
```

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```
..\ITEfolder3\file1.txt
..\ITEfolder3\file2.txt
2 File(s) copied
```

```
C:\Users\TIPQC\.BossBlazo>xcopy ..\.BossBlazo3
..\.BossBlazo3\file1.txt
..\.BossBlazo3\file2.txt
2 File(s) copied
```

c. At the prompt, type **dir** to display the content of **ITEfolder1**. Only the files in the **ITEfolder3** were copied into **ITEfolder1**. The directory **ITEfolder4** was not copied into **ITEfolder3**.

```
C:\Users\ITEUser\ITEfolder1> dir
<some output omitted>
 Directory of C:\Users\ITEUser\ITEfolder1
04/29/2019  03:16 PM    <DIR>          .
04/29/2019  03:16 PM    <DIR>          ..
04/29/2019  03:05 PM                20 file1.txt
04/29/2019  03:05 PM                20 file2.txt
               2 File(s)             40 bytes
               2 Dir(s)  31,491,321,856 bytes free
```

```
C:\Users\TIPQC\.BossBlazo>dir
 Volume in drive C is Q3203-16
 Volume Serial Number is 7029-55C0

 Directory of C:\Users\TIPQC\.BossBlazo

10/08/2026  12:54 PM    <DIR>          .
10/08/2026  11:41 AM    <DIR>          ..
10/08/2026  12:08 PM                19 doc1.txt
10/08/2026  12:12 PM                19 doc2_copy.txt
10/08/2026  12:12 PM                19 file.log
10/08/2026  12:12 PM                20 file1.txt
10/08/2026  12:12 PM                20 file2.txt
               5 File(s)             97 bytes
               2 Dir(s)  519,313,059,840 bytes free
```

d. Use **help xcopy** or **xcopy /?** to determine which switch would allow the **xcopy** command to copy **all** the files and directories.

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Question:

What option allows you to copy all the files and directories, including the empty directories?

```
C:\Users\TIPQC\.BossBlazo>help xcopy
Copies files and directory trees.

XCOPY source [destination] [/A | /M] [/D[:date]] [/P] [/S [/E]] [/V] [/W]
                           [/C] [/I] [/-I] [/Q] [/F] [/L] [/G] [/H] [/R] [/T]
                           [/U] [/K] [/N] [/O] [/X] [/Y] [/-Y] [/Z] [/B] [/J]
                           [/EXCLUDE:file1[+file2][+file3]...] [/COMPRESS]
                           [/[-]SPARSE] [/NOCLONE]

  source       Specifies the file(s) to copy.
  destination  Specifies the location and/or name of new files.
  /A           Copies only files with the archive attribute set,
               doesn't change the attribute.
  /M           Copies only files with the archive attribute set,
               turns off the archive attribute.
  /D:m-d-y     Copies files changed on or after the specified date.
               If no date is given, copies only those files whose
               source time is newer than the destination time.
  /EXCLUDE:file1[+file2][+file3]...
               Specifies a list of files containing strings.  Each string
               should be in a separate line in the files.  When any of the
               strings match any part of the absolute path of the file to be
               copied, that file will be excluded from being copied.  For
               example, specifying a string like \obj\ or .obj will exclude
               all files underneath the directory obj or all files with the
               .obj extension respectively.
  /P           Prompts you before creating each destination file.
  /S           Copies directories and subdirectories except empty ones.
  /E           Copies directories and subdirectories, including empty ones.
               Same as /S /E. May be used to modify /T.
  /V           Verifies the size of each new file.
  /W           Prompts you to press a key before copying.
  /C           Continues copying even if errors occur.
  /I           If destination does not exist and copying more than one file,
               assumes that destination must be a directory.
  /-I          If destination does not exist and copying a single specified file,
               assumes that destination must be a file.
  /Q           Does not display file names while copying.
  /F           Displays full source and destination file names while copying.
  /L           Displays files that would be copied.
  /G           Allows the copying of encrypted files to destination that does
               not support encryption.
  /H           Copies hidden and system files also.
  /R           Overwrites read-only files.
  /T           Creates directory structure, but does not copy files. Does not
               include empty directories or subdirectories. /T /E includes
               empty directories and subdirectories.
  /U           Copies only files that already exist in destination.
  /K           Copies attributes. Normal Xcopy will reset read-only attributes.
  /N           Copies using the generated short names.
  /O           Copies file ownership and ACL information.
  /X           Copies file audit settings (implies /O).
  /Y           Suppresses prompting to confirm you want to overwrite an
               existing destination file.
  /-Y          Causes prompting to confirm you want to overwrite an
               existing destination file.
  /Z           Copies networked files in restartable mode.
  /B           Copies the Symbolic Link itself versus the target of the link.
  /J           Copies using unbuffered I/O. Recommended for very large files.
  /COMPRESS    Request network compression during file transfer where
               applicable.
  /[-]SPARSE   Enable or disable retaining the sparse state of files during
               copy. /-SPARSE overrides /SPARSE if both are provided.
  /NOCLONE     Do not attempt block cloning as an optimization.
```

e. Because **ITEfolder4** is a subfolder and **ITEfolder5** is both a subfolder and an empty folder, **/E** is needed to copy all the contents of **ITEfolder3** and the empty subfolder.

Type **xcopy /E ..\ITEfolder3 .** at the prompt to copy the files. When prompted, type **a** to allow overwriting the existing files.

```
c:\Users\ITEUser\ITEfolder1> xcopy /E ..\ITEfolder3
Overwrite C:\Users\ITEUser\ITEfolder1\file1.txt (Yes/No/All)? a
..\ITEfolder3\file1.txt
..\ITEfolder3\file2.txt
2 File(s) copied
```

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```
C:\Users\TIPQC\.BossBlazo>xcopy /E ..\.BossBlazo3
Overwrite C:\Users\TIPQC\.BossBlazo\file1.txt (Yes/No/All)? A
..\.BossBlazo3\file1.txt
..\.BossBlazo3\file2.txt
2 File(s) copied
```

f. Verify that the **ITEfolder4** and **ITEfolder5** were also copied in **ITEfolder1**.

```
c:\Users\ITEUser\ITEfolder1> dir
<some output omitted>
 Directory of c:\Users\ITEUser\ITEfolder1

04/29/2019  04:41 PM    <DIR>          .
04/29/2019  04:41 PM    <DIR>          ..
04/29/2019  03:05 PM                20 file1.txt
04/29/2019  03:05 PM                20 file2.txt
04/29/2019  04:41 PM    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  31,493,193,728 bytes free

c:\Users\ITEUser\ITEfolder1> dir ITEfolder4
<some output omitted>
 Directory of c:\Users\ITEUser\ITEfolder1\ITEfolder4

04/29/2019  04:41 PM    <DIR>          .
04/29/2019  04:41 PM    <DIR>          ..
04/29/2019  03:00 PM    <DIR>          ITEfolder5
               0 File(s)              0 bytes
               3 Dir(s)  31,493,193,728 bytes free
```

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```
C:\Users\TIPQC\.BossBlazo>dir
 Volume in drive C is Q3203-16
 Volume Serial Number is 7029-55C0

 Directory of C:\Users\TIPQC\.BossBlazo

10/08/2026  12:56 PM    <DIR>          .
10/08/2026  11:41 AM    <DIR>          ..
10/08/2026  12:56 PM    <DIR>          .BossBlazo4
10/08/2026  12:08 PM                19 doc1.txt
10/08/2026  12:12 PM                19 doc2_copy.txt
10/08/2026  12:12 PM                19 file.log
10/08/2026  12:12 PM                20 file1.txt
10/08/2026  12:12 PM                20 file2.txt
               5 File(s)             97 bytes
               3 Dir(s)  519,073,513,472 bytes free

C:\Users\TIPQC\.BossBlazo>dir .BossBlazo4
 Volume in drive C is Q3203-16
 Volume Serial Number is 7029-55C0

 Directory of C:\Users\TIPQC\.BossBlazo\.BossBlazo4

10/08/2026  12:56 PM    <DIR>          .
10/08/2026  12:56 PM    <DIR>          ..
10/08/2026  11:52 AM    <DIR>          .BossBlazo5
               0 File(s)              0 bytes
               3 Dir(s)  519,070,367,744 bytes free
```

g. The robocopy command can also be used to copy the content of a directory to a new destination. The robocopy command has more capabilities than the built-in Windows copy and xcopy command, such as resume copying after a network interruption, skip files that appear to be identical to the files in the destination folders, and mirror a directory by keeping the destination directory in sync with the source directory.

Copy the ITEfolder4 content to ITEUser folder using the robocopy command.

```
C:\Users\ITEUser\ITEfolder1> robocopy /E
C:\Users\ITEUser\ITEfolder3\ITEfolder4\ C:\Users\ITEUser
```

Note the information provided by the command during the copying process.

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```
C:\Users\TIPQC\.BossBlazo>robocopy /E

------------------------------------------------------------------------------
   ROBOCOPY     ::     Robust File Copy for Windows
------------------------------------------------------------------------------

  Started : Thursday, October 8, 2026 12:59:17 PM
   Source -
     Dest -

    Files : *.*

  Options : *.* /S /E /DCOPY:DA /COPY:DAT /R:1000000 /W:30

------------------------------------------------------------------------------

ERROR : No Source Directory Specified.

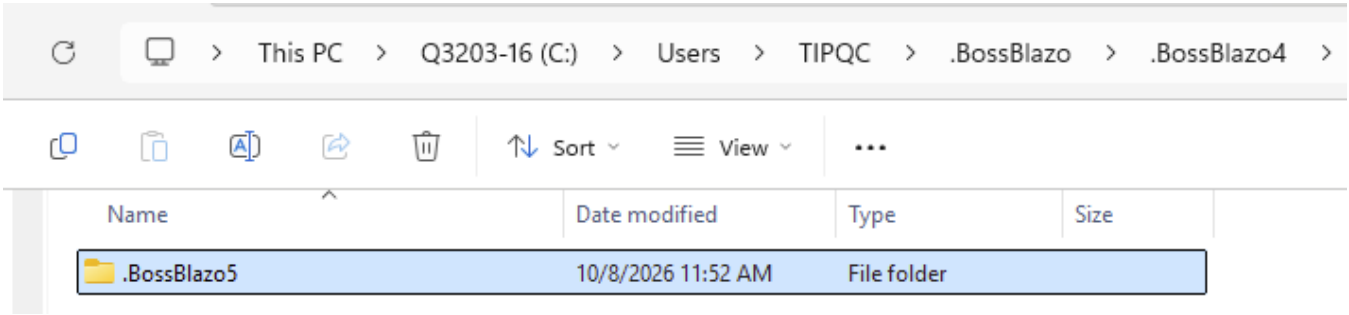
       Simple Usage :: ROBOCOPY source destination /MIR

            source :: Source Directory (drive:\path or \\server\share\path).
         destination :: Destination Dir  (drive:\path or \\server\share\path).
               /MIR :: Mirror a complete directory tree.

    For more usage information run ROBOCOPY /?


****  /MIR can DELETE files as well as copy them !
```

h. Navigate to **C:\Users\ITEUser** to verify that the folder ITEfolder5 is copied.



Step 6: Delete directories.

In this step, you will delete an empty and a non-empty directory using the **rd** command.

a. Use the **rd ITEfolder2** to delete the empty directory and verify that the directory was deleted.

```
C:\Users\ITEUser> rd ITEfolder2
```

```
C:\Users\TIPQC>rd .BossBlazo2
```

b. Navigate to the **C:\Users\ITEUser\ITEfolder3** directory.

c. Use the **rd ITEfolder1** to delete the directory. Verify the directory removal using the **dir** command.

Were you able to delete the directory? Explain.

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```
C:\Users\TIPQC>rd .BossBlazo

C:\Users\TIPQC>dir
 Volume in drive C is Q3203-16
 Volume Serial Number is 7029-55C0

 Directory of C:\Users\TIPQC

10/08/2026  01:15 PM    <DIR>          .
03/09/2026  11:03 AM    <DIR>          ..
07/30/2025  06:05 PM    <DIR>          .anaconda
01/21/2025  10:55 AM    <DIR>          .azuredatastudio
10/08/2026  12:39 PM    <DIR>          .BossBlazo3
10/02/2025  04:44 PM    <DIR>          .cache
09/24/2025  05:22 PM    <DIR>          .conda
07/30/2025  06:05 PM                25 .condarc
07/30/2025  06:05 PM    <DIR>          .continuum
08/01/2026  03:08 PM    <DIR>          .copilot
07/30/2025  06:10 PM    <DIR>          .ipython
07/30/2025  06:13 PM    <DIR>          .jupyter
07/30/2025  06:10 PM    <DIR>          .matplotlib
09/03/2026  12:09 PM               176 .packettracer
07/18/2026  02:35 PM    <DIR>          .spyder-py3
10/08/2026  12:32 PM    <DIR>          .VirtualBox
08/06/2025  05:06 PM    <DIR>          .vscode
07/18/2026  04:41 PM    <DIR>          .vscode-shared
09/03/2026  01:33 PM    <DIR>          Cisco Packet Tracer 8.2.2
10/18/2024  05:21 AM    <DIR>          Contacts
04/28/2026  09:15 AM    <DIR>          Desktop
09/18/2026  06:27 PM    <DIR>          Documents
10/08/2026  11:05 AM    <DIR>          Downloads
05/02/2025  08:23 AM    <DIR>          Favorites
02/10/2026  10:59 AM    <DIR>          flutter
08/06/2025  06:50 PM             2,361 from PyQt6.py
07/30/2025  06:34 PM    <DIR>          Jedi
10/18/2024  05:21 AM    <DIR>          Links
10/07/2026  05:26 PM    <DIR>          Microsoft
10/18/2024  05:21 AM    <DIR>          Music
10/18/2024  05:23 AM    <DIR>          OneDrive
09/16/2026  02:38 PM    <DIR>          Pictures
08/01/2026  03:09 PM    <DIR>          PyQt6
10/18/2024  05:21 AM    <DIR>          Saved Games
10/17/2024  03:58 PM    <DIR>          Searches
08/06/2025  05:09 PM    <DIR>          venv
04/14/2026  09:59 AM    <DIR>          VHDPlus
05/21/2026  02:10 PM    <DIR>          Videos
10/08/2026  11:20 AM    <DIR>          VirtualBox VMs
               3 File(s)          2,562 bytes
              36 Dir(s)  518,299,443,200 bytes free
```

Yes using rd prompt

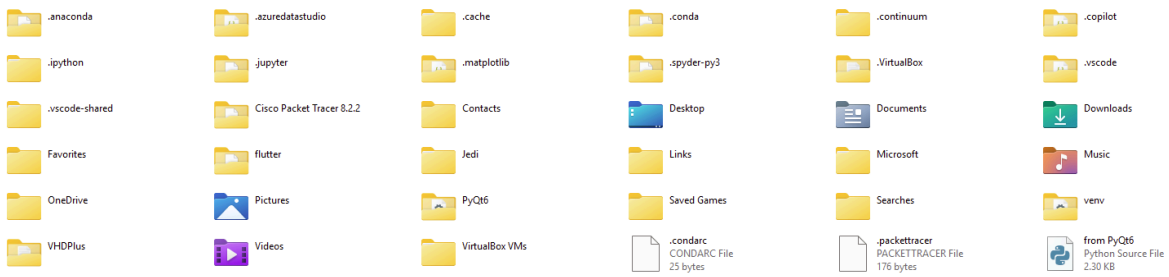
d. Use **rd /?** command to determine the switch that allows the deletion of a non-empty directory.

```
c:\Users\ITEUser\ITEfolder3> rd /S ITEfolder1
ITEfolder4, Are you sure (Y/N)? Y
```

```
C:\Users\TIPQC>rd /S .BossBlazo
.BossBlazo, Are you sure (Y/N)? Y
```

e. Use the appropriate commands to delete all the text files and folders that you have created in this lab. Type **exit** to close the command prompt window.

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Reflection Question

What are the advantages of using CLI vs. GUI?

The CLI is faster and uses fewer system resources than a GUI. It gives users more control and allows them to perform tasks using commands and scripts. CLI is also useful for troubleshooting and automating repetitive tasks.

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